

From Microscope Photographs to Battery Physics: An Interpretable, Uncertainty-Aware Pipeline for Silicon–Graphite Anode Cross-Sections

clean-room edition — only what the data support

micro2dfn pipeline — technical report

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Headline

- **No detectable composition difference.** Candidate-silicon fraction 6.0%/6.1%/6.0% (B1/B2/B3, medians); resolved pore fraction 10.4%/10.8%/10.9%. At $n = 7$ per batch, differences under ~ 2 area-points are invisible.
- **No batch difference is separable from imaging session.** The photographs fall into 13 distinguishable acquisition groups; the only unblocked signals are count-type markers that the data itself flags as session-dominated, and every one collapses under within-session blocking.
- **Two Batch_1 fields exceed the reference bright range** (12.5%, 11.9% vs. reference max 10.7%); a third sits marginally above it (10.7%). Real as an image observation; unresolved as material — in the one shared session it matches a Batch_2 field.
- **Next steps, in order:** acquisition metadata or interleaved re-imaging (breaks the session–batch tangle); EDS (names the bright material); measured porosity (widest model error bar).

Contents

1	The measurement system	2
2	Primary markers: area fractions	2
3	The acquisition confound	2
4	What was demoted, and why	3
5	What was removed, and why	4
6	Limitations	4
A	DFN consequence indicators (demoted)	4
B	Exploratory channel features	5
C	Provenance	5

1 The measurement system

Thirty-one BSE cross-section images (7 Batch_1, 7 Batch_2, 17 Batch_3 reference), ~ 25 nm/px, plus three unlabelled new images. The pipeline: flat-field correction $\rightarrow$ three-class segmentation (pore / graphite-bulk / bright material) using thresholds fitted *once on the reference batch only* ($t_{\text{pore}} = 0.711$, $t_{\text{Si}} = 1.335$, $t_{\text{core}} = 1.550$), saved in `recipe.json` and reused verbatim — new images can never move the ruler. A per-object classifier then separates bright objects into *candidate-silicon* and *uncertain-bright* (interior-brightness and compactness rules calibrated on the same reference). `run_manifest.json` records recipe, code and image hashes for every run.

What “reproducible” means here. The recipe is deterministic: two independent pipelines fit the same thresholds to four decimal places. That demonstrates *repeatability of the ruler*, not its validity — stability against real specimen and imaging variation still requires re-imaging and ground truth.

Resolution floors. Pores below ~ 20 px (≈ 0.13 μm) and silicon objects below ~ 150 px (≈ 0.35 μm) are absorbed into bulk — sub-resolution porosity and fines are invisible by design.

The labels, honestly. “Bright” is a BSE-contrast label, not a chemical one: silicon, silicon-oxide-rich phases and some artefacts all read bright. “Candidate silicon” means *passes the silicon shape and brightness rules*; “uncertain bright” means *bright but fails them* — not “artefact” and not “silicon”. Chemical identity is an EDS question.

Objects extracted: 31,515 bright objects — 12,197 classified candidate-silicon, 19,318 uncertain-bright.

2 Primary markers: area fractions

The primary comparison uses **area fractions only**. Section 3 shows why: count-type markers track the imaging session far more closely than the material; area fractions are merge-invariant (a pore that blurs into two objects keeps the same area) and are the least session-sensitive marker class.

Table 1: Batch medians for the primary fraction markers.

Fraction	Batch_1	Batch_2	Batch_3
Resolved pore	10.4%	10.8%	10.9%
Candidate silicon	6.0%	6.1%	6.0%
Uncertain bright	1.6%	2.0%	1.9%
Total bright	7.7%	8.4%	7.7%

Composition overlaps completely: no pair difference in these four markers reaches significance even before correction. The correct statement is *no detectable difference* — gaps under about two area-points are beyond reach at $n = 7$ per batch, and equal candidate fractions do not establish equal chemistry.

The three bright Batch_1 fields. Against the reference maximum (10.7%), two Batch_1 photos sit clearly above (12.5%, 11.9%) and a third marginally above (10.7%). In both flagged low-contrast photos, most of that excess is *uncertain* bright material (classified silicon reads only 3.0% and 2.6%). And in the one acquisition session shared with Batch_2, the third field’s bright fraction is nearly identical to its same-session partner (10.7% vs. 10.3%). Solid as an observation; “can’t tell” as a batch claim.

3 The acquisition confound

Thirteen groups, inferred not recorded. Grouping the 31 photos by frame size, pixel size, detectors, noise and contrast yields 13 acquisition groups — inferred imaging sessions (recorded metadata does not yet exist; request filed). The groups predict every acquisition property almost perfectly and cut across batch labels: the reference batch alone spans seven of them; five groups contain more than one batch.

How we test against them. Each marker’s variance is partitioned after removing group means versus batch means; lower residual = that variable explains more. With 13 groups on 31 images the *chance level* — the residual pure noise would show — is ≈ 0.60 after group and ≈ 0.93 after batch; both must be quoted with the numbers.

Table 2: Variance partition on the declared marker family (chance levels: 0.60 after group, 0.93 after batch). The pattern, not any single number, is the result: count-type markers are the most session-organized class; fractions the least — though groups still explain a substantial share of every marker.

Marker	After group	After batch	Type
pore count / Mpx	0.04	0.53	count
pore count (noise-adj)	0.06	0.62	count
pore median size	0.06	0.43	boundary/size
Si count	0.31	0.80	count
uncertain-bright frac.	0.06	0.84	fraction
total bright frac.	0.19	0.86	fraction
candidate-Si frac.	0.21	0.89	fraction
Si clustering R	0.22	0.93	boundary/size
pore frac.	0.23	0.95	fraction
Si median size	0.30	0.93	boundary/size
Si 90% size	0.65	0.95	boundary/size

Why counts are fragile. A count depends on where object boundaries land; acquisition contrast and sharpness shift them, merging or splitting objects while conserving total area. Even a minimum-size cut-off does not rescue the count — the session effect is edge drift across the whole size range, not specks at the detection floor. The pipeline’s working rule: *fractions are primary; counts are within-session diagnostics only*.

Unblocked comparison (declared family, exact permutation + Benjamini–Hochberg + minimum detectable difference). Of 42 declared tests, 5 reach nominal $p < 0.05$ (2.1 expected by chance), and 3 survive BH at $q = 0.10$ — all count-type markers on the Batch_1–Batch_3 pair: pore count +34.5/Mpx ($p_{\text{adj}}=0.025$), noise-adjusted pore count +18.8/Mpx ($p_{\text{adj}}=0.029$), median pore size $-0.028\ \mu\text{m}$ ($p_{\text{adj}}=0.039$). Every survivor belongs to the session-dominated marker class — self-consistently so.

Blocked comparison (stratified permutation, batch labels shuffled within groups only). Only five groups mix batches, and shared coverage is thin: Batch_1–Batch_2 share 3 sessions, Batch_2–Batch_3 share 3, and **Batch_1–Batch_3 share exactly one session with one photo each** — the decisive pair is nearly untestable. Within that limit: **zero** of the 42 blocked tests reach nominal significance, including on the markers that survived unblocked. In the single session shared by the key pair, Batch_1 has *fewer* pores than Batch_3 (123 vs. 141/Mpx).

Does the pipeline see material or sessions? A classifier given only acquisition information (frame height, noise, contrast) reaches 68% leave-one-out batch accuracy, above the 55% majority baseline; the full image-feature set reaches 42%. The imaging conditions carry more apparent batch information than the material features. The three new-image calls are consistent with this: every photo’s frame height occurs in only one batch, and each call followed the session — the predictions were saved blind, but the outcome cannot distinguish batch-match from session-match. All three are reported “can’t tell”.

Honest boundary. This does not prove the batches identical — the within-session test is under-powered. It proves the dataset cannot separate material from imaging condition, and that no count-based or unblocked batch claim is supportable as measured evidence.

4 What was demoted, and why

- **Boundary-dependent markers** (particle sizes and shapes, chords, anisotropy, clustering index, crack fraction, percolation reach, largest pore region, correlation lengths): every batch comparison on them is nominal-or-less unblocked and zero blocked; they also depend on where boundaries land, which the session moves. Appendix material.
- **Silicon-access markers** (distance to pore, enclosed share, coverage, accessible fraction): built on one 2-D slice over resolved pores only — 2-D sees $\sim 10\%$ porosity of a real 25–40%, and out-of-plane contacts are invisible. Kept as stated limitations inputs to the model, not as batch markers.
- **The InLens/ETD layer** (exploratory): InLens gain differs by batch (median brightness ≈ 89 for Batch_3 vs. ~ 110 for Batches 1–2), so even its within-session-consistent direction (Batch_1 pores darkest in all 3 shared sessions) rests on $n = 3$. Promising, unproven — follow-up with controlled re-imaging.

- **New-image assignments:** kept blind protocol; all three calls “can’t tell” pending session control.
- **DFN indicators** (Appendix A): model-conditional, assumption-dominated, session-inheriting.

5 What was removed, and why

- **FFT “vertical texture” and the curtaining test:** the directionality score measured the frame’s aspect ratio, not the texture — pure noise scores the same as real images at the real frame shape; direct autocorrelation shows features slightly *horizontally* elongated. Archived.
- **Diffusion-solve tortuosity:** no 2-D section ever has a spanning pore path, so the solve returned $+\infty$ everywhere — it never produced a usable number. The honest residual is one Limitations sentence.
- **Swelling-deficit as a finding:** negative by construction (resolved porosity $\sim 10\%$ vs real 25–40%) and “deficit tiles track silicon” true by definition. The real content — silicon wetting depends on pores we cannot see — stays in Limitations as a blind spot.
- **3-D reconstruction branch** (optional SliceGAN): not ground truth, not needed for the core deliverable; archived runnable.

6 Limitations

- **Acquisition and material are not adequately separated** — the dominant limitation. Groups are inferred; shared sessions are thin (B1–B3: one). “Can’t tell” is the correct status for every batch difference, not “identical”.
- **“Bright” $\neq$ “silicon”** — EDS is the gate.
- **Resolved porosity $\approx 10\%$ vs true 25–40%** — silicon wetting runs through sub-resolution pores we cannot see; a real blind spot, stated not hidden.
- **Absolute model values are illustrative;** all assumed parameters (thickness, sub-resolution porosity, mechanics) are swept and still cap accuracy.
- **No 3-D connectivity** from 2-D sections.
- **BSE cannot see** binder, SEI, lithium inventory, wetting, foil corrosion.

A DFN consequence indicators (demoted)

The Doyle–Fuller–Newman model translates image markers into cell responses; it adds no measurement of its own. Under the repaired model (in-cycle peaks, one shared cathode per sweep index, corrected plating rule), paired 17-point sweeps give: cycle-5 discharge capacity 5.59/5.68/5.48 Ah (B1/B2/B3); peak swelling $\sim 12\ \mu\text{m}$; peak silicon surface stress 10.0/9.5/10.8 MPa. Only the stress ordering exceeds its declared threshold (B2–B3 = $-1.60\ \text{MPa}$, -15% of the B3 value; B1–B3 = -0.95 ; B1–B2 = $+0.53$), driven almost entirely by the *modelled* reachable-silicon share (Spearman $\rho \approx -0.93$) — a 2-D contact proxy, not measured connectivity. Assumptions alone move capacity by 1.14 Ah against $\leq 0.15\ \text{Ah}$ of batch difference, and the label-uncertainty bracket moves stress by up to 1.8 MPa — more than the batch difference itself. A further caveat found in this audit: the saved sweeps fed particle-radius overrides under invalid PyBaMM key names, so both radii ran at their parameter-set defaults for every batch and point — the radius channel contributed nothing to the saved numbers (fixed in code for the next run). Read as a hypothesis for the image layer, not a batch verdict; the model also inherits the session confound through its inputs. Lithium loss, retention and efficiency are diagnostics only (nothing ages measurably in 5 cycles).

B Exploratory channel features

The co-registered InLens and ETD detector channels (0-px shift on all 34 fields) answer different physical questions than BSE; they are recorded and analysed in `channel_features.csv` but kept out of the frozen ruler. Within-session-consistent direction: Batch_1's pores read darkest in InLens in all 3 shared sessions — a candidate follow-up signal, not a measured pore depth, with $n = 3$ and batch-correlated gain.

C Provenance

Recipe `recipe.json` (baseline Batch_3, $t_{\text{pore}} = 0.711$, $t_{\text{Si}} = 1.335$, $t_{\text{core}} = 1.550$); every run writes `run_manifest.json` (recipe, code and image hashes). All numbers in this paper are generated by `build_report.py` from `per_image_features.csv`, `image_acquisition.csv`, `objects_classified.csv` and `dfn_paired_differences.csv`. `check_report.py` fails on any stale or hand-typed value.